Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Data Engineer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production web and data applications for clients via Upwork and Fractal, specializing in streaming data pipelines, event-driven architecture, and automated testing.
- **Order Procurement & Management Solution** | *Kafka, PostgreSQL, Python, TypeScript*
  - Architected a full-stack order portal using NextJS, ExpressJS, TypeScript, and PostgreSQL with RESTful APIs, session authentication, and automated audit logging.
  - Engineered an automated ELT ingestion pipeline parsing WhatsApp messaging streams into structured PostgreSQL database records, reducing order processing latency to <150ms.
  - Modeled relational PostgreSQL database schemas with custom indexing and integrity constraints, enforcing schema validation and data contracts for 2,000+ daily order transactions.
  - Streamed order event logs from Kafka streams into a Snowflake warehouse via Python, embedding shift-left data quality checks at the source for real-time operational analytics.

Software Developer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Flink, C/C++, Django, Python* **(Live link)**
  - Built a native C/C++ bridge using `dart:ffi` to port OpenCV Scale-Invariant Feature Transform (SIFT) keypoint algorithms into Flutter for single-codebase cross-platform vision processing.
  - Constructed a low-latency video stream pipeline normalizing raw camera feed buffers into C structs with zero memory leaks, maintaining 30+ FPS throughput under peak processing loads.
  - Implemented stateful stream processing logic for keypoint extraction, transmitting match coordinates asynchronously to Dart with <33ms frame-processing latency for overlay rendering.
  - Developed RESTful backend APIs in Django with SQL query optimization to power location-aware spatial discovery features, transforming raw spatial event streams for alpha release.
- **Kia National Conference App** | *Python, Flutter, PostgreSQL*
  - Automated a real-time event data pipeline syncing registration records for 500+ attendees at the Kia National Dealer Conference in Bangkok, applying CI/CD practices for deployments.

Technical Skills

Languages & Core Systems: Python, SQL (Expert), C/C++, TypeScript, JavaScript, C, Dart
Data Engineering & Streaming: Kafka, Flink, Confluent Schema Registry, Data Mesh, DataOps, ETL/ELT, Data Contracts
Databases & Cloud Warehouses: PostgreSQL, Snowflake, AWS, Django, ExpressJS, NodeJS, NextJS, RESTful APIs
Developer Tools & Infrastructure: Docker, CI/CD, Git, Linux, Bash, LaTeX, OpenCV (SIFT), NumPy, Pandas

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)